

ϕ TD-Gaussian

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.06141	0.14	0.1202	0.0842
Cramér $\ell_2^2 \downarrow$	0.0006766	0.002284	0.002927	0.001618
CF - $\ell_2^2/\omega^2 \downarrow$	0.0005699	0.001558	0.00256	0.001342

ϕ TD-Cauchy

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1197	0.03785	0.07609	0.1322
Cramér $\ell_2^2 \downarrow$	0.003941	0.0001825	0.001764	0.002499
CF - $\ell_2^2/\omega^2 \downarrow$	0.002608	0.0001576	0.0008532	0.001347

ϕ TD-Gamma

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.7164	2.58	1.955	0.1258
Cramér $\ell_2^2 \downarrow$	0.2984	0.8119	0.7406	0.001809
CF - $\ell_2^2/\omega^2 \downarrow$	0.2627	0.7464	0.7005	0.001861

ϕ TD-Categorical

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2809	0.4091	0.3342	0.3069
Cramér $\ell_2^2 \downarrow$	0.02767	0.02058	0.03749	0.02875
CF - $\ell_2^2/\omega^2 \downarrow$	0.02087	0.01651	0.03084	0.02309

ϕ TD-Quantile

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1017	0.2176	0.1389	0.1868
Cramér $\ell_2^2 \downarrow$	0.002228	0.003044	0.004003	0.004065
CF - $\ell_2^2/\omega^2 \downarrow$	0.001678	0.003617	0.003402	0.003629